SC2006 SOFTWARE ENGINEERING

Lecture 12: System Design

Responsibilities, interactions, and design trade-offs

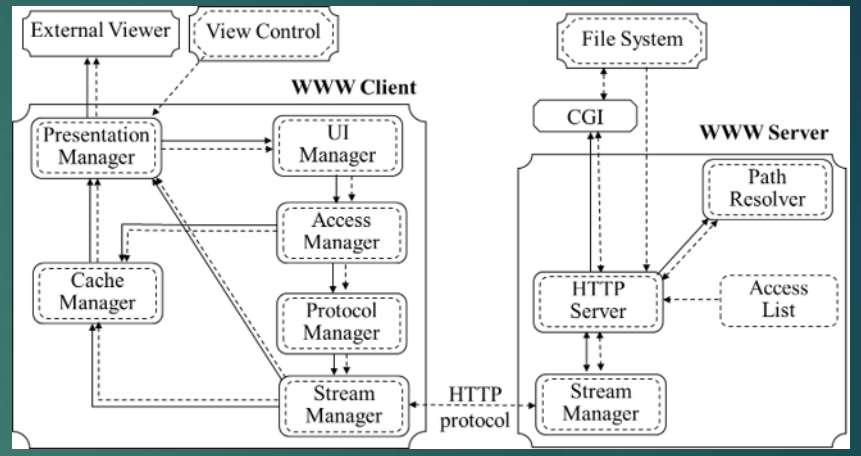
Liu Yang

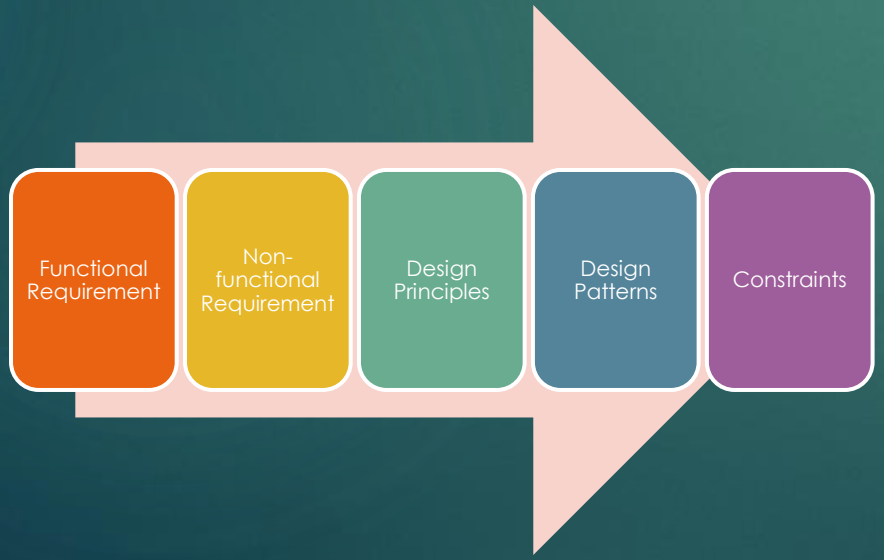
Recap of System Design

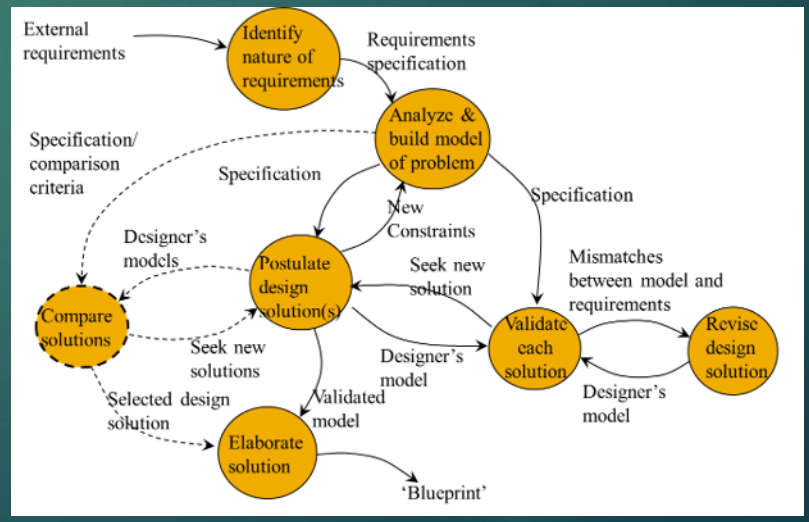
Why

What

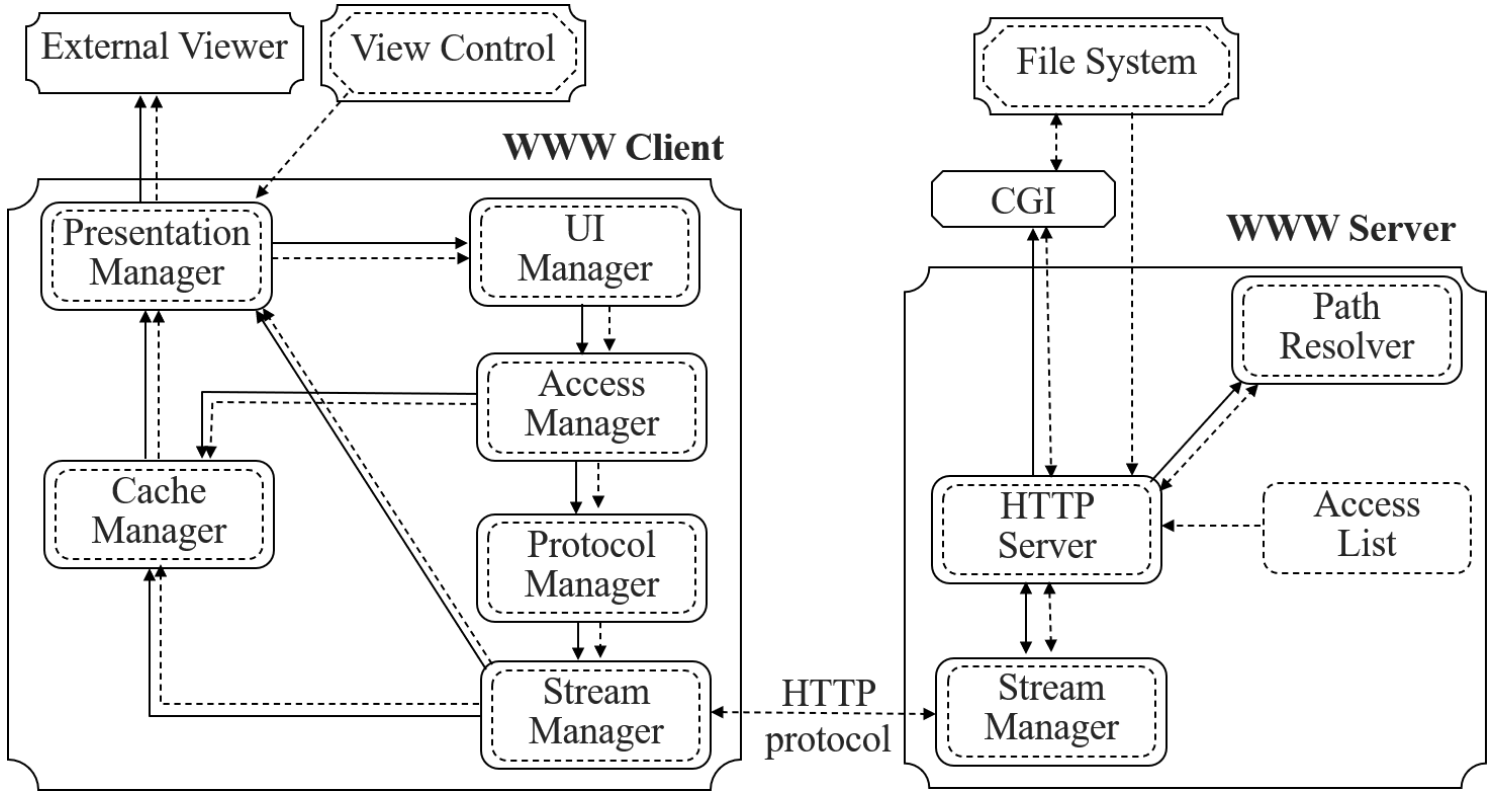
How

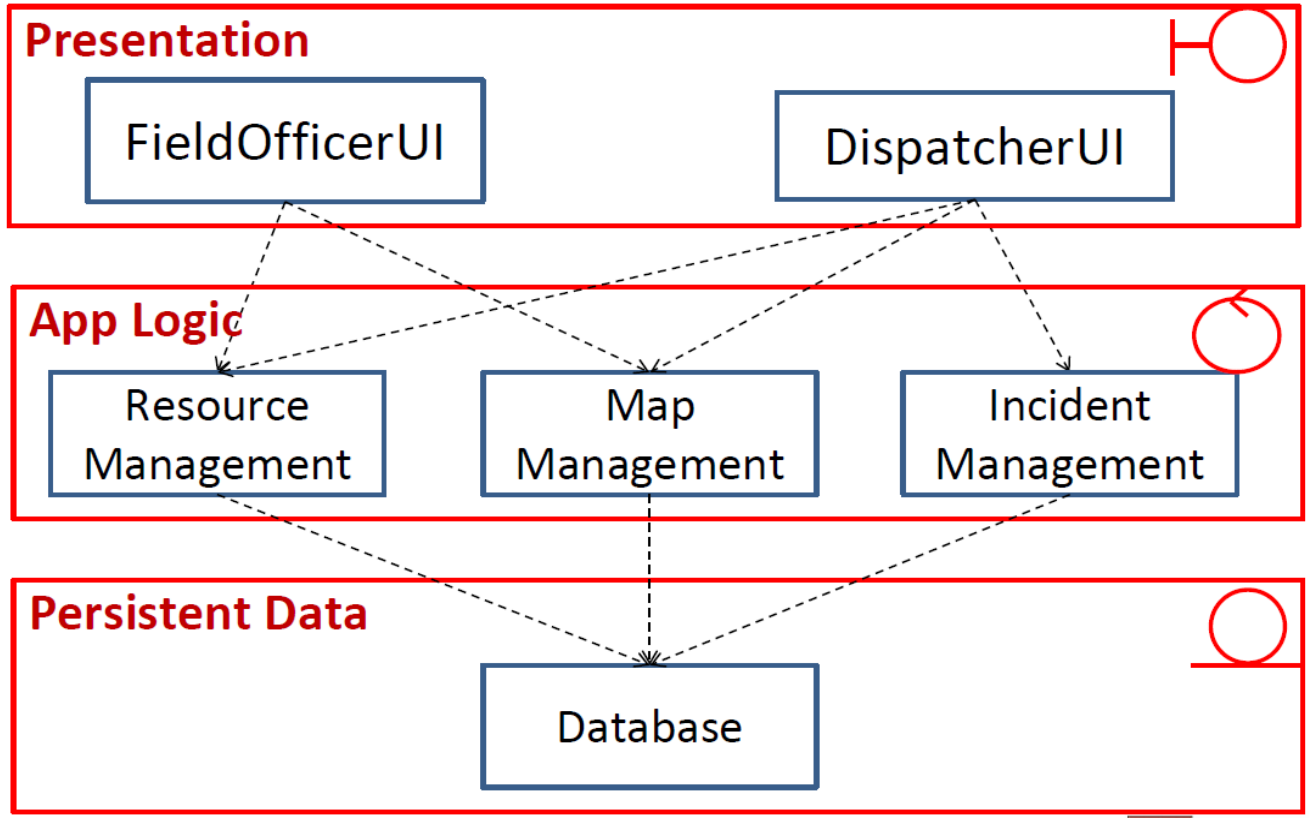




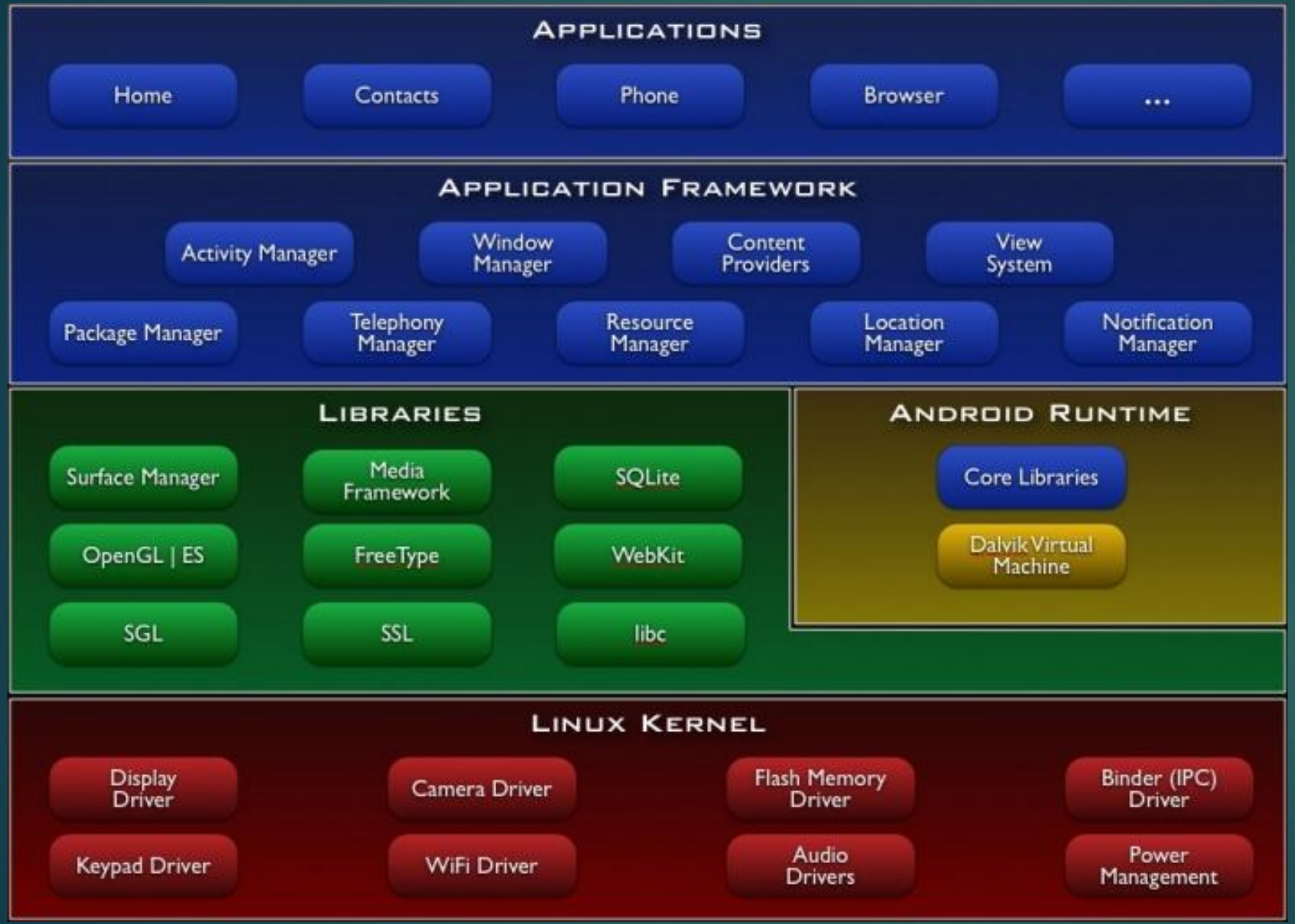


WWW Client-Server Architecture





Layered Architecture – 3-Layered Architecture



Examples: Android OS

Analysis of Layered Structure Style

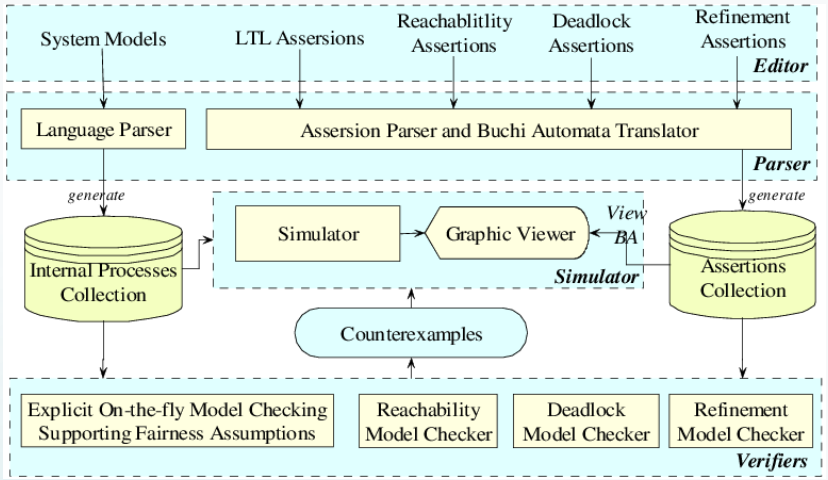
Advantages

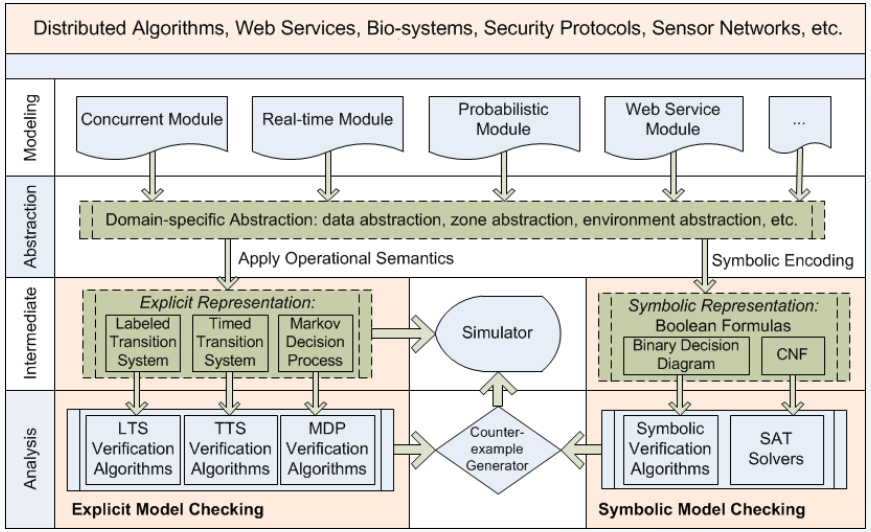
- Support designs based on increasing levels of abstraction
- Support enhancement of functionality by introducing new components
- Support reuse of lower layer components while upper layer components varies

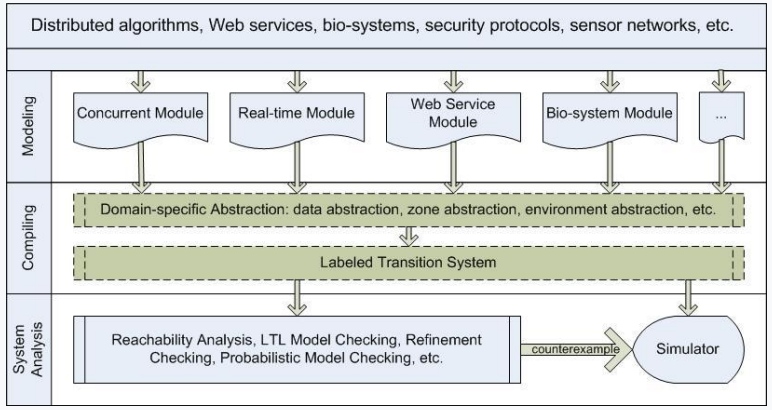
Disadvantages

- Not all systems are easily structured in a layered fashion
- Difficult to find the right levels of abstraction
- Lower performance

Example of Architecture Evolution





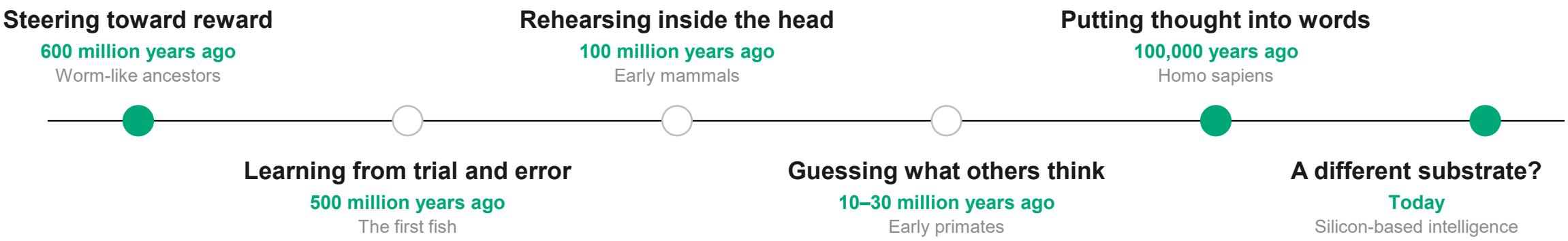


What is the Intelligence?

TIMELINE OF INTELLIGENCE

Four Billion Years of Intelligence, Five Turning Points

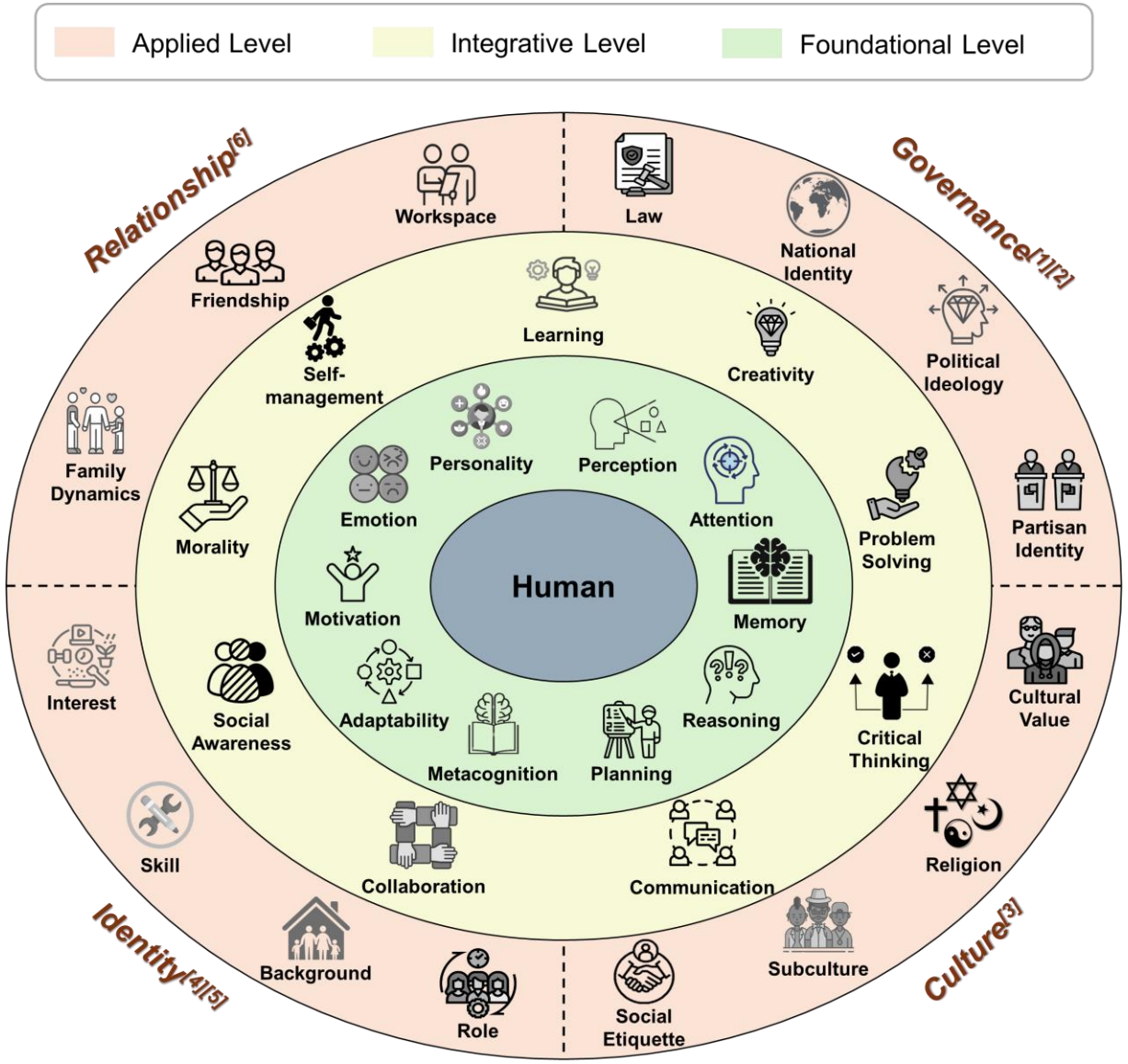
Every breakthrough was a by-product of the one before it.



The book's terms, in order: Steering · Reinforcing · Simulating · Mentalising · Speaking

**Language sits at the very end of this line —
yet it carries almost all of human civilisation.**

Each new ability appeared not because the brain grew larger,
but because the old strategy could no longer hold the new pressure.

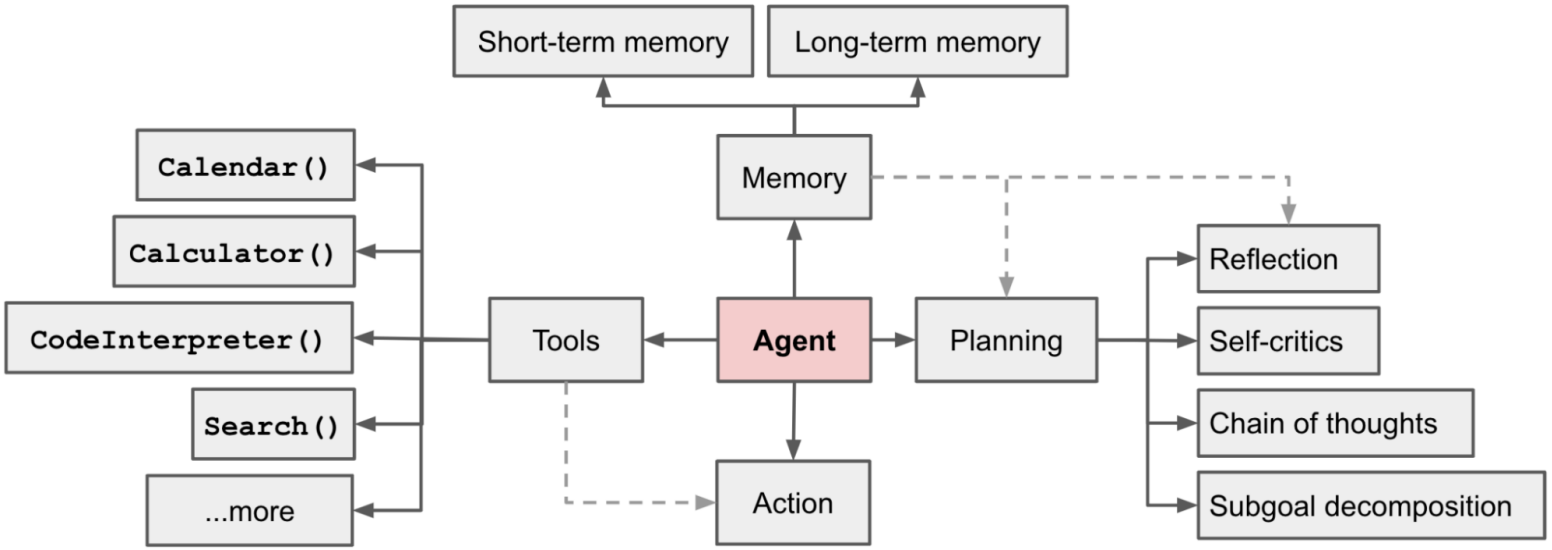


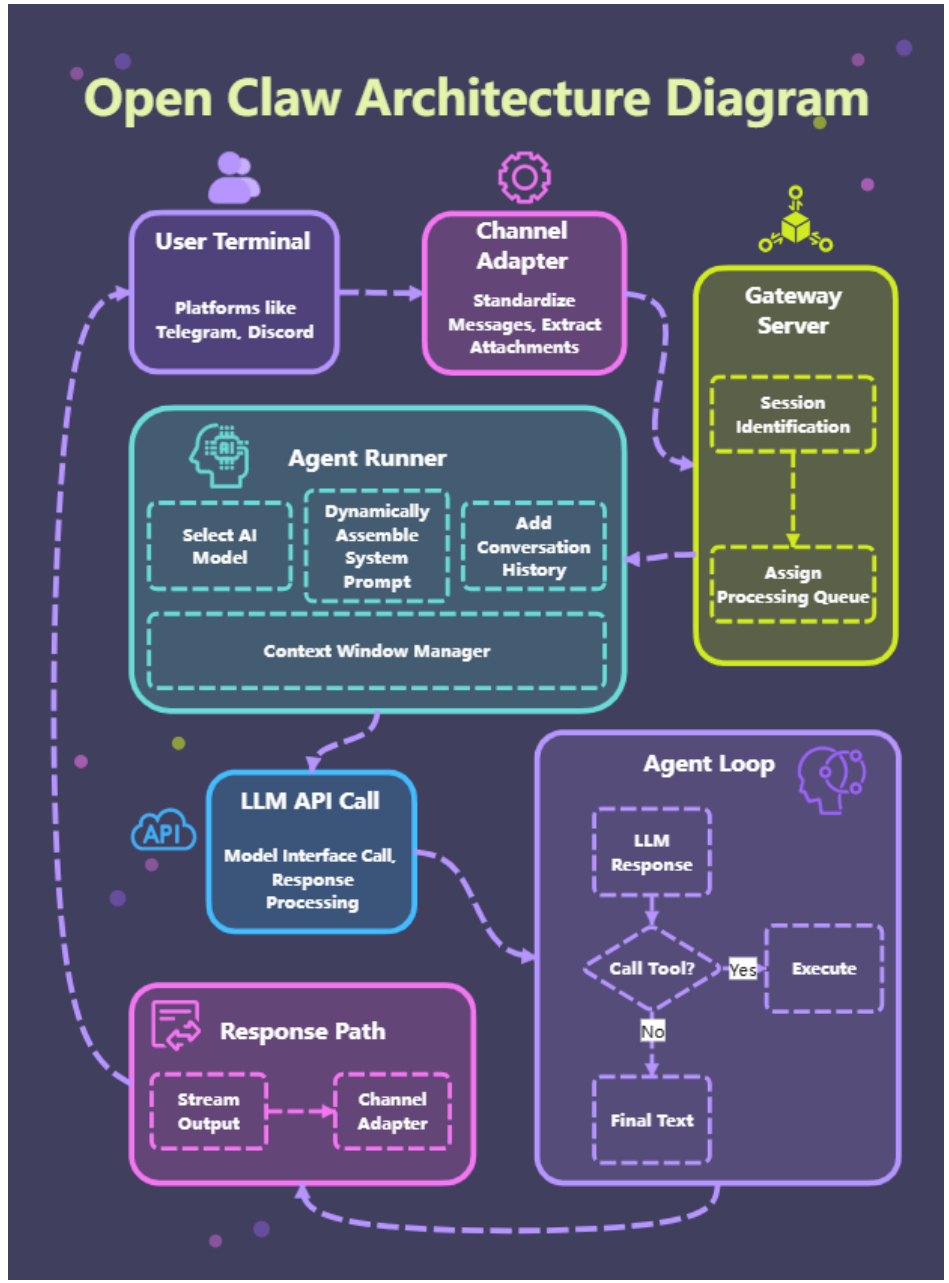
Three-tier Capability System

- Foundational Level
  - ***Basic capabilities*** that supports individuals' cognition and behaviors.
  - Refer to execution functions and intelligence theory [1-3].
- Integrative Level
  - Build upon basic capabilities, ***the key for individuals to cope with the complex and rapidly*** developing modern society.
  - Refer to the 21st century competencies theory [4-6].
- Applied Level
  - ***High-level capabilities related to application scenarios,*** based on the foundational and integrative levels.
  - Refer to cognitive science theory [7-12].

[1] Beyond IQ: A triarchic theory of human intelligence. 1985.
[2] Mainstream science on intelligence: An editorial with 52 signatories, history, and bibliography. 1997.
[3] The unity and diversity of executive functions and their contributions to complex "frontal lobe" tasks: A latent variable analysis. Cognitive Psychology. 2020
[4] Framework for 21st Century Student Outcomes. 2019.
[5] Comparing frameworks for 21st century skills. 2010.
[6] 21st century skills: What are they and how do we assess them? Applied measurement in education. 2016.
[7] AI for identifying social norm violation. Scientific Reports, 2023.
[8] Political ideology: Its structure, functions, and elective affinities. Annual Review of Psychology, 2009.
[9] Culture and the self: Implications for cognition, emotion, and motivation. Psychological Review, 1991
[10] An integrative theory of intergroup conflict. 1979.
[11] The psychology of life stories[J]. Review of general psychology, 2001.
[12] The strength of weak ties. American journal of sociology, 1973.

Architecture of LLM Powered Autonomous Agents





- https://lilianweng.github.io/posts/2023-06-23-agent/

Codex Architecture: 135 Crates in Total

From bottom to top: gradually abstracted, and more modular, extensible, and pluggable

L5 · User Interface Layer

codex-cli
Provides 27 commands, including: analyze, etc.

L4 · Frontend / Service Layer

codex-tui
238,882 lines

app-server-client
Service communication and orchestration

codex-exec
Non-interactive execution

codex-app-server
128,440 lines

codex-mcp-server
4,128 lines

CI builds: no direct dependency on codex-core

L3 · Intelligent Core

codex-core-plugins
37,043 lines

codex-core
298,779 lines · **67** subdirectories
Core component

codex-tools
6,777 lines

exdl / 12 × crate
(8 crates with Rust API)

L2 · Protocols and Agreements

extension-api

codex-protocol
71 crates in total
Core component

app-server-protocol
30,971 lines

exec-server-protocol

L1 · Capability Layer

config · login · http-client · sandboxing · state · otel

L0 · Common Utilities

utils/ 24 crates — among them, utils-absolute-path (55 crates) depend on the crate utils

L1 and L0 are all crates located under the same directory, and the entire project is organized in a flat structure.
The red dashed line indicates the CI build dependency: the "direct dependency" from codex-tui to codex-core is not present.
The "direct import" dependency refers to the dependency relationship between crates within the same crate (i.e., codex-tui and codex-core are in the same crate).

The Agent's Three-Layer Loop: One Sentence In, Three Nested Loops Inside the Core

Layer 1 · Submission Loop `submission_loop` — Receive Submission from the channel and dispatch by Op (26 actions)

Layer 2 · Task SessionTask — Four categories: normal conversation / context compression / code review / user_shell

Layer 3 · turn Loop — One 'round of conversation' you actually see

run_turn(...)
Assemble context → call model → use tools → write history

Any queued input?

No

Return result
Task complete

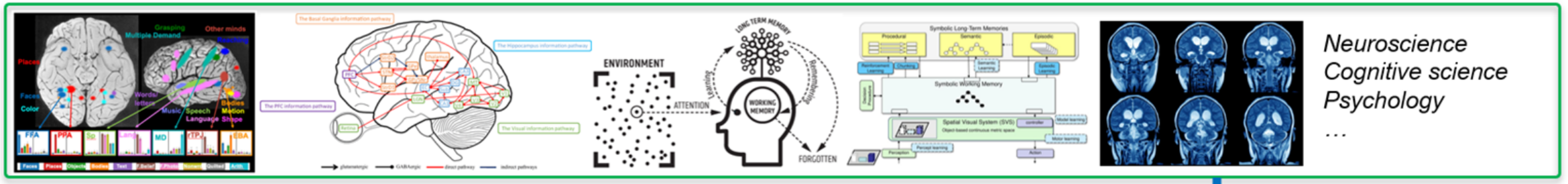
Yes → do not return; immediately run another round

This loop is the implementation principle behind 'you can keep typing while the model is still running'

There are only two external entry/exit points for the session:
Commands enter via `tx_sub` → events exit via `rx_event` (about 80 `EventMsg` types)

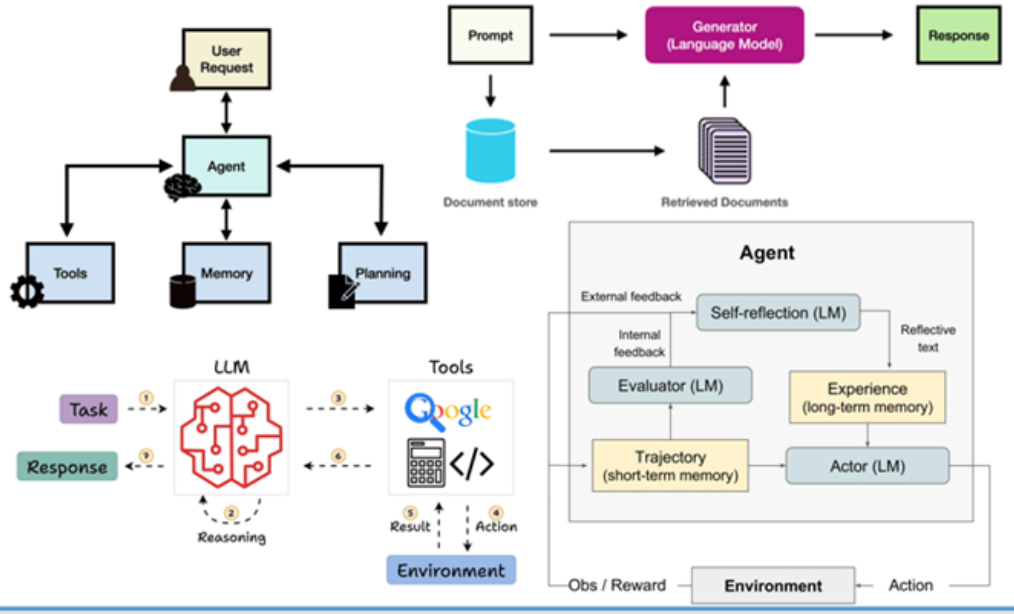
Each layer answers a question:
Layer 1 "Did anything come in?" Layer 2 "What kind of work is this?"
Layer 3 "Is this round finished?"

Brain-Inspired Agent Computation Architecture?



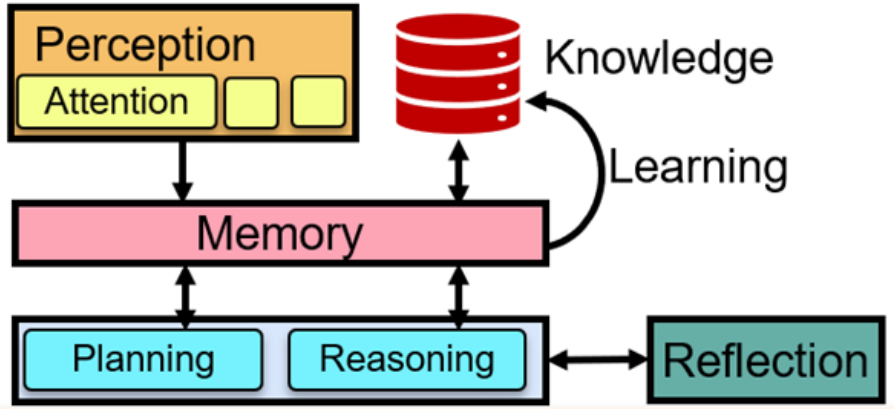
guidance

Current Agent Architectures



evolution

Brain-Inspired Agent Architecture



- Works like a brain
- Thinks like a brain
- Adapts like a brain
- Brain-like robust
- Brain-like intelligence

Brain-Inspired Agent Archecture

“All definite knowledge belongs to science; all dogma as to what surpasses definite knowledge belongs to theology.

But between theology and science there is a No Man’s Land, exposed to attack from both sides; this No Man’s Land is philosophy.”

Bertrand Russell

